

INTELLIGENCE & EXECUTION

ShadowTalk AI AI Engines & Agentic Tools

Deep dive into the Turbo Engine complexity routing, sovereign on-device WebGPU runtimes, 30+ native tools, cognitive loops, and adaptive memory.

ROUTING ENGINE

Turbo Engine (Groq + OpenAI + Claude)

TOOL REGISTRY

30+ Autonomous Tool Handlers

ON-DEVICE AI

WebGPU + WebLLM Quantized Gemma

MEMORY ARCHITECTURE

IndexedDB Vectorized Adaptive Memory

1 The Turbo Engine: Dynamic Model Routing

Dispatching every user inquiry to heavy frontier models introduces unnecessary token expenditure and unacceptable latency for simple greetings. Conversely, routing complex system design questions to lightweight models leads to reasoning failures. ShadowTalk solves this through the **Turbo Engine** in `src/lib/turbo/`.

Heuristic Complexity Scoring (`analyzeComplexity`)

Every message array is evaluated before dispatch based on semantic keywords and character payload volume:

- **Low Complexity:** Matches common greetings (`hi`, `hello`, `thanks`), formatting instructions (`tldr`, `fix typo`), or prompt lengths under 100 characters. Automatically routed to **Groq** (`llama-3.1-8b-instant`), achieving token generation speeds exceeding 300 tokens/sec.
- **Medium Complexity:** General conversational turns, creative synthesis, and standard queries routed to **Groq** (`llama-3.3-70b-versatile`).
- **High Complexity:** Triggered by architectural terms (`refactor`, `architect`, `security`, `typescript`, `sql`, `synthesize`) or contexts exceeding 4,000 characters. Escalated to **OpenAI** (`gpt-4o`) or **Anthropic** (`claude-3-5-sonnet`).

Sovereign & On-Device Inference

ShadowTalk provides full support for air-gapped and zero-leak operations:

- **WebGPU Local Runtime:** `src/lib/webgpuRuntime.ts` probes device VRAM. On compatible GPUs, models like Gemma-2B run client-side using WebAssembly and WebGPU compute shaders via `@mlc-ai/web-llm`.
- **Ollama Desktop Bridge:** When running locally, ShadowTalk communicates with local daemon endpoints on `http://localhost:11434` without passing data to external servers.
- **BYOK (Bring Your Own Key):** Stored strictly in local encrypted storage, authorizing direct edge client-to-provider streaming.

2 The 30+ Native Tool Registry

The tool orchestration pipeline (`src/hooks/useToolOrchestrator.ts` and `executeShadowTool.ts`) provides conversational execution of native actions:

Tool Identifier	Target Handler	Capability Description
<code>web_search</code>	Cloud Function <code>web-search</code>	Queries live web indexes, retrieves snippets, and synthesizes multi-source citations.
<code>deep_research</code>	Turbo Engine Prompt Graph	Executes multi-step recursive web research, formulating follow-up queries and final reports.
<code>image_generator</code>	Pollinations / Flux API	Generates photorealistic images with custom seeds, aspect ratios, and styles.
<code>image_decoder</code>	Multimodal Vision Bridge	Performs OCR, document inspection, and diagram reasoning on base64 image attachments.
<code>shadow_browser</code>	Firecrawl Scraping Gateway	Fetches live JS-rendered URLs, stripping ads and converting DOM to clean Markdown.
<code>code_executor</code>	WebContainer Node.js Engine	Spawns processes, executes scripts in sandboxed WebAssembly, and streams terminal output.
<code>presentation_builder</code>	Client <code>pptxgenjs</code> Pipeline	Creates PowerPoint slides, bullet points, styling, and exports directly to <code>.pptx</code> .
<code>database_query</code>	Backend Function <code>postgres-query</code>	Runs schema inspection and sanitized SQL read queries, formatting rows into JSON tables.
<code>shadowspectre</code>	Cyber Defense Agent	Performs vulnerability assessments, security header audits, and exploit path validation.

3 Cognitive Loops & Swarm Consensus

Beyond single-turn tool calls, ShadowTalk implements multi-step autonomous cognitive loops:

The Cognitive Loop Framework (CognitiveLoopPanel.tsx)

When activated, the agent enters an autonomous 4-stage execution cycle:

1. **Plan:** Breaks high-level user directives into structured, ordered sub-tasks.
2. **Execute:** Sequentially dispatches sub-tasks to tools or reasoning models.
3. **Reflect:** Evaluates tool execution output against the original user criteria. If an error or partial result occurs, the agent self-corrects.
4. **Synthesize:** Blends all sub-task artifacts into an exhaustive final output.

Multi-Model Consensus (useMultiModelConsensus.ts)

For high-stakes decision making, ShadowTalk broadcasts the identical prompt simultaneously across three distinct model families (e.g. Groq Llama-3.3, OpenAI GPT-4o, and Google Gemini 2.0). The responses are analyzed for semantic convergence, surfacing agreements, edge-case discrepancies, and a calculated confidence rating.

4 Adaptive Memory Engine

Persistent user context is managed by `src/lib/memory/adaptiveMemory.ts`. Unlike dumb chat transcripts, Adaptive Memory scores and categorizes high-signal statements:

Turn Scoring Heuristics

- **Identity Signals:** Statements matching `my name is`, `i am`, `i build` gain +0.40 confidence.
- **Preference Signals:** Matches on `i prefer`, `always`, `never` gain +0.35 confidence.
- **Decision Signals:** Matches on `we decided`, `going with` gain +0.30 confidence.
- **Task Signals:** Matches on `build`, `ship`, `deploy` gain +0.20 confidence.

IndexedDB Storage Schema

- **Store Name:** `facts` in database `shadowtalk_memory_db`.
- **Indexes:** Indexed on `by-confidence` and `by-lastUsedAt`.
- **Auto-Pruning:** When facts exceed capacity, least-recently-used (LRU) low-confidence facts are purged to avoid context bloat.